Assignment #4 for Numerical Analysis

Deadline: Mar. 14

1. (40 points) Use Theorem 1.11 or 1.12 to estimate the error e_{i+1} in terms of the previous error e_i as Newton's Method converges to the given roots. Is the convergence linear or quadratic?
   (a) $2x^4 - 5x^3 + 3x^2 + x - 1 = 0; r = -1/2, r = 1$
   (b) $x^3 - x^2 - 5x - 3 = 0; r = -1, r = 3$
2. (10 points) Prove that Newton's Method applied to $f(x) = ax + b$ converges in one step.
3. (10 points) Apply two steps of Newton's Method with initial guess
   (a) $x^3 + x - 2 = 0, x_0 = 0$
   (b) $x^2 + \frac{1}{x+1} - 3x = 0, x_0 = 1$
4. (30 points) Consider Newton's Method for calculating $\sqrt{a}$.
   (a) Show that the Newton's Method formula is given by
   $$x_{k+1} = \frac{1}{2}(x_k + \frac{a}{x_k}), x_0 > 0.$$
   (b) Prove that $x_k \geq \sqrt{a}$ for $k = 1, 2, \cdots$, and the sequence $\{x_1, x_2, \cdots\}$ is decreasing.
5. (10 points) Sketch a function f and initial guess for which Newton's Method diverges.